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Locating a Missing Football from Player Information

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Contents

Abstract **v**

1 Introduction **1**
1.1 Background . . . 1
1.2 Problem and purpose . . . 1
1.3 Research question and objectives . . . 2
1.4 Contribution and scope . . . 3
1.5 Report structure . . . 3

2 Research Methodology **4**
2.1 Study approach . . . 4
2.2 Football images and the training set . . . 4
2.3 Label assignment and comparison methods . . . 5
2.4 Prediction measurements . . . 6
2.5 Supporting checks . . . 7
2.6 Paired comparisons and reliability . . . 8

3 Literature Review **9**
3.1 Prediction of a missing object . . . 9
3.2 Ball detection and tracking . . . 9
3.3 Gaze direction and image locations . . . 10
3.4 Gaze following from scene information . . . 10
3.5 Joint attention and player pose . . . 11
3.6 Player selection and location models . . . 11
3.7 Football data and image editing . . . 12
3.8 Evaluation and the role of simple guesses . . . 12

4 System Design **14**
4.1 Purpose and processing path . . . 14
4.2 The earlier head and eye approach . . . 15
4.3 Image preparation . . . 15
4.4 Player detection . . . 16
4.5 Player features . . . 16
4.6 Player selection . . . 17
4.7 Predicting the ball position . . . 18
4.8 Raw and blended predictions . . . 18
4.9 Controls and saved outputs . . . 19

5 System Evaluation **20**
5.1 The comparison . . . 20
5.2 Results on the two samples . . . 20
5.2.1 The original 61 images . . . 20

5.3 Contribution of the player features . . . 22
5.4 Close estimates and larger errors . . . 22
5.5 Understanding the raw model's errors . . . 25
5.5.1 Training-label eligibility . . . 25
5.5.2 Player selection . . . 25
5.5.3 What the offsets add . . . 26
5.6 Supporting checks . . . 27
5.6.1 Inputs and replay . . . 27
5.6.2 An additional distant edit . . . 27
5.6.3 Older person-marked predictions . . . 27
5.6.4 Earlier head-and-eye heatmaps . . . 28
5.7 Discussion . . . 28
5.8 Limitations . . . 29

6 Conclusion **30**
6.1 Answer to the research question . . . 30
6.2 Contribution of the work . . . 30
6.3 Next steps . . . 31

A Implementation and supporting results **32**
A.1 Settings needed to repeat the comparison . . . 32
A.2 Original paired comparisons . . . 33
A.3 Details of the follow-up checks . . . 34
A.4 The common 49-image comparison . . . 34
A.5 Data use . . . 35
A.6 Supplementary player-selection comparisons . . . 35

List of Figures

4.1 Main player-based prediction path, read from top to bottom. The system uses the training-position mean when no player remains after filtering. 14

5.1 Distribution of point errors 23
5.2 Coordinate examples 24

List of Tables

2.1 Training set and the original and larger evaluations. 5

5.1 Point-localisation results 21
5.2 Results on all images from the same matches 22
5.3 Results by training-label eligibility 25
5.4 Answer-assisted selection diagnostic 26
5.5 Supplementary raw and player-centre comparisons 26

A.1 Original paired error differences 33
A.2 Expanded paired error differences 33
A.3 Raw prediction paired comparisons 34
A.4 Answer-assisted paired changes 34
A.5 Matched secondary comparison 35
A.6 Supplementary paired mean-error differences 35
A.7 Supplementary paired close-rate differences 36

Abstract

Players positions and body poses may offer clues when a football cannot be seen in a photograph, this project developed a tool to predict its location from those cues. Balls were digitally removed from images with their recorded positions, allowing each prediction from an edited image to be checked against the balls actual position.

Existing detection and pose models provided player positions and estimated body landmarks, such as shoulders and knees. These measurements and the known ball positions were used to train separate models to select a player and predict the ball's position relative to that player. The raw prediction was then blended with the average training-ball position to test whether this reduced large errors.

The training set contained 3,238 images from 318 matches. Evaluation first used 61 images from 61 other matches, then all 813 available images from those same evaluation matches. The player-based predictions were compared with the image centre, the average training-ball position and Gaze-LLE, a pretrained gaze-target model.

The raw predictions included close estimates alongside large misses, so blending reduced average error but also moved some good predictions away from the ball. On the 61-image sample, the blended system did not clearly improve on centre guessing, while the 813-image comparison supported a minor improvement. Gaze-LLE was more accurate overall. Further checks found a minor benefit from learned offsets compared with using the selected player's centre, and much larger errors when no detected player was near the ball. The project contributes the player-based method and an evaluation showing how player selection, relative location prediction and blending affect its performance.

Chapter 1

Introduction

1.1 Background

A football image shows multiple clues about the balls location. Players may turn towards it, move to receive it or gather around the same part of the pitch, so their positions and body poses may still offer useful clues when the ball cannot be seen. This project investigates whether a computer can use those clues to predict the missing ball's location.

To test this idea, the study uses still images from football broadcasts in which the ball's position has already been recorded. The ball is digitally removed while the players and surrounding scene remain visible, and the tool predicts where it had been. That prediction is then compared with the recorded position to measure the error. This missing-ball task builds on earlier research [1].

A ball detector searches for the ball's appearance. Once that appearance has been removed, a prediction has to depend on other information in the image. Players are one possible source because they form part of the action around the ball. Their positions and body arrangements can be recorded and compared across images, which provides the starting measurements for the tool.

1.2 Problem and purpose

The project asks whether measurements taken from the players improve the predicted ball location. The tool finds people in the image, records their positions, sizes and estimated body points, such as shoulders and knees, then selects one player as the reference for its prediction. This gives the location models one set of measurements to work with. The relationships are learned from training images with known ball positions, while evaluation uses edited images without the true ball position.

The player-based estimates are compared with two simple baselines: the image centre and the average ball position from the training images. Both return a fixed point for every input, providing a reference for whether the player measurements improve prediction.

A scene-based model provides a useful comparison because it can draw on the surrounding image as well as the players. Gaze-LLE is a model trained to estimate where a person is looking [2], and its authors found that it could suggest a gaze target without a marker selecting the person's head. In scenes with several people, a marker was needed to identify whose gaze to estimate. Here, the unmarked setting avoids a separate head-selection step and allows a proposal even when no usable head is detected. Gaze-LLE receives the edited image, and the highest-scoring location in its heatmap becomes the ball-position estimate.

By suggesting a location to inspect, such a tool might help someone checking or annotating an image where the football is difficult to see. The aim is to measure how close that suggestion is to the recorded ball position.

1.3 Research question and objectives

The research question is:

> How useful is information about players for locating a missing football, compared with simple position guesses and a scene-based model?

The study has four objectives:

1. Build a player-based system that predicts a ball location from an image with the ball removed.

2. Prepare and check the images and labels, keeping the true ball positions out of the prediction inputs.

3. Compare the system with simple position references and Gaze-LLE on the same evaluation cases.

4. Examine the contribution of the player measurements and explain the main sources of prediction error.

Average error describes how far the system tends to miss, while the number of close predictions shows how often it gives a useful estimate under a stated distance rule. A method can get close in several images and still have a poor average because its other mistakes are much larger.

1.4 Contribution and scope

The project provides a working player-based prediction system and tests its main choices. Its player features describe position, the size and shape of the detection box, and estimated body landmarks. The evaluation examines whether these features help, whether predicting a distance from the selected player improves on using that player's centre, and what happens when predictions are pulled towards a typical ball position.

Development began with head and eye direction, but the limited results led to predicting ball locations from player positions and body points. The thesis focuses on this player-based approach and uses the earlier direction estimates to explain how it developed.

Scoring uses the physical ball's labelled position. The data do not record which player was looking at it, so the findings concern missing-ball localisation in single images.

1.5 Report structure

Chapter 2 sets out the comparison and its measurements. Chapter 3 reviews the research behind the selected approaches. Chapter 4 describes how the tool turns an image into a ball prediction. Chapter 5 presents the results and explains the patterns of error. Chapter 6 answers the research question and identifies the most useful next steps.

Chapter 2

Research Methodology

2.1 Study approach

The study compares ways of locating a football after it has been removed from an image. Player measurements and known ball positions are used to train models that select a player and predict the ball's position relative to that player. The predictions are checked against the original positions, alongside simple baselines and a pretrained scene model evaluated on the same images.

After the initial head-and-eye work, development moved to learning ball positions from player measurements. The comparison used data, settings and measurements recorded before image preparation and training, starting with one image per match and later adding images to any particular sources of error.

All 379 matches in the local collection had already been used during earlier development, making this an exploratory study. Evaluation matches were excluded from the current training set to prevent direct training overlap in this comparison [3], although that separation does not remove the earlier exposure.

2.2 Football images and the training set

The study uses 4,051 images from 379 matches, a subset of the football broadcast images and ball annotations provided by SoccerNet-v3D [4]. Each image has a labelled box around the football, whose centre gives the physical location used for scoring. The labels describe the ball location, with no eye-tracking measurements for individual players.

The original evaluation uses 61 images, one from each of 61 matches in the source test partition. Selection followed a rule based on match and frame identifiers, without considering the ball location, visible faces or model predictions. One frame per match gives each match equal influence in this first comparison. As

nearby frames can share players, camera views and moments of play, the training set excludes every image from these matches.

The larger evaluation adds the other 752 available images from the excluded matches, which gives 813 images from the same 61 matches. Table 2.1 summarises this division.

Table 2.1: Training set and the original and larger evaluations.

Use	Images	Matches
Training	3,238	318
Original evaluation	61	61
Additional images in the follow-up	752	Same 61
Complete follow-up evaluation	813	Same 61

Image identities, dimensions and ball coordinates were checked against the source records before training. Two groups of repeated source images were found and kept in their original training groups, with no copies crossing into evaluation.

Both training and evaluation use edited images, with Telea inpainting filling the ball area from the surrounding pixels [5]. The labelled ball position is used to prepare the images and train the location models, then kept out of normal prediction, which receives only the prepared image and player features. Scoring compares the output with the true ball position. Because removal can leave traces, the evaluation also checks whether an edit could influence the scene model.

2.3 Label assignment and comparison methods

Training needs labels for which player to select and how far the ball lies from that player. Since the ball labels do not identify where players are looking, the detected player box nearest the ball supplies a consistent reference without manually labelling player roles. Distance is zero when the ball point is inside the box. Otherwise, it is measured to the nearest edge. A frame qualifies when this distance is at most 0.05 of the image diagonal, measured in pixels. The cutoff came from the earlier implementation and was kept without optimisation in this comparison. The location models learn the left/right and up/down distances from that player's box centre to the ball, but prediction still runs on all evaluation images, including those where the ball is far from any player.

Of the 3,238 training images, 1,957 qualified under this rule, containing 18,540 player rows. Each row contains one detected player's measurements, and all rows train the selector, with the assigned player marked as the choice in each frame. The location models use only the assigned player's measurements, giving 1,957 examples of the distance to the ball. The training-position mean is calculated

separately from the ball coordinates across all training images and supplies the fallback whenever detection and filtering have no player to select.

The full system uses player position, box shape and body landmarks: estimated points such as shoulders and knees. Its *raw prediction* is the point calculated directly from the selected player's measurements. The main method, called the *full blend*, combines one third of that point with two thirds of the training-position mean. This weight was from earlier exploratory development and kept unchanged for this comparison. Pulling the prediction towards a typical ball location limits extreme estimates but can also move good estimates away from the true ball position. Because the raw point is kept inside the image first, blending bounds the output to the central rectangle described in Chapter 4. Reporting both predictions shows the effect of this restriction.

Two further methods examine what the extra measurements contribute. The *position-only blend* uses player boxes, detection scores and a score for nearby green pitch, without body landmarks. The *shuffled-description blend* keeps each player's position but exchanges their extra box-shape and body measurements with another player in the same image. A fixed shuffle for each image makes this check repeatable while changing both player selection and location prediction.

The *training-position mean* and *image centre* are simple guesses that do not change with the input image. Centre predicts $(0.5, 0.5)$ and needs no training data. The training mean uses the average labelled location in the training set. Because balls may tend to appear in particular parts of a broadcast frame, a learned system could obtain a reasonable score without making much use of the players. Comparing it with these guesses checks how much the current image contributes.

Gaze-LLE completes the comparison [2]. It receives the same edited scenes without a supplied head marker identifying one person. Its highest-scoring heatmap location becomes the point prediction. Appendix A records this conversion. The output is tested as an image-based ball estimate, providing a reference without the new tool's separate player-selection stage. Older person-marked Gaze-LLE outputs are assessed separately on the common 49 images, because outputs are missing for the other 12.

All seven methods are scored on every selected image, including fallback cases, so the comparison includes inputs where player detection fails.

2.4 Prediction measurements

Scoring divides horizontal coordinates by image width and vertical coordinates by image height, so both axes run from zero to one. The distance between prediction

$(\hat{x}_i, \hat{y}_i)$ and the true ball position (x_i, y_i) is

$$e_i = \sqrt{(\hat{x}_i - x_i)^2 + (\hat{y}_i - y_i)^2}. \tag{2.1}$$

The primary score is the full blend's mean distance across the original 61 images, with lower values indicating smaller errors. Each image has equal weight, so large mistakes can noticeably raise the average. The median helps show how these larger mistakes affect the result.

A prediction is reported as *close* when this distance is at most 0.10. On a $1,920 \times 1,080$ image, a purely horizontal miss of 192 pixels or a purely vertical miss of 108 pixels reaches that threshold. This scale differs from the label-assignment rule's pixel-diagonal distance. The predictor returns a point rather than a probability region.

The planned main comparisons are between the full blend and the training mean, position-only blend, centre and Gaze-LLE. Differences are calculated by subtracting the comparison method's error from the full blend's error, so a negative value favours the full blend. Individual improvements and worsening predictions are also recorded to show which cases fall behind the averages.

2.5 Supporting checks

The eligibility check applies the same nearest-box rule to the evaluation labels and reports all seven methods for qualifying and non-qualifying images. This asks whether images resembling the training examples are easier for the models. Both groups remain in the main comparison.

The answer-assisted diagnostic, defined here as selection using the ground-truth ball position, selects the nearest player before applying the full location models and blending rule. It uses the usual fallback for empty detections and tests the same kind of player association used during training. A nearby box may still be an unhelpful reference, and this label describes proximity rather than where a player was looking.

A supplementary check compares the full raw estimate with raw position-only prediction and the full selector's chosen player centre. Under answer-assisted selection, it also compares that player's centre with the offset prediction. These comparisons use saved predictions, with samples, fallbacks and scoring unchanged. Appendix A.6 gives the construction and paired comparisons.

The editing check adds an inpainted patch to see whether it attracts the scene prediction. A recorded random procedure places a patch matching the original removal mask's dimensions at a distant, non-overlapping location. Its centre must be at least 0.20 from the ball in the separately normalised image coordinates. Rerunning the original inputs first reproduced all 61 native scene maps exactly.

The comparison measures the change in ball error, distance to the patch and movement towards it.

2.6 Paired comparisons and reliability

Paired bootstrap resampling describes how differences vary across the observed matches [6]. Each of 5,000 draws samples whole matches with replacement, and every method receives the same sampled matches. Each comparison has a central pointwise 95% percentile interval from the resulting differences. If it crosses zero, the data do not clearly establish which method has the lower error. The saved analysis includes all 21 pairs among the seven methods and all 28 pairs in the eight-method comparison on 49 images.

The main summary averages errors and close rates within each match before averaging those match results. All 61 matches therefore have equal influence despite their different frame counts. Separate image-weighted summaries describe the images collectively. Each resample includes every frame from each selected match because images from the same match are related.

Saved inputs and outputs link each score to its image, player features and trained models. Separate calculation and replay code checked the predictions, distances and intervals without training new models.

Chapter 3

Literature Review

3.1 Prediction of a missing object

Once a football has been removed, its location has to be inferred from what remains in the image: the players' positions, body shapes and surrounding scene. Research on missing objects examines how reliably those clues can identify a location.

In *Spot the Ball*, humans and vision-language models predict missing ball locations on an overlaid grid in football, basketball and volleyball images [1]. Balamurugan et al. report errors from defaulting to the image centre and choosing the wrong player as a reference. Both are relevant to a player-based method, making simple position guesses useful comparisons. The football evaluation here uses continuous coordinates from recorded ball boxes, with different images, editing and scoring, so its scores cannot be compared directly with the benchmark.

3.2 Ball detection and tracking

Tracking offers a way to estimate a ball's location when it disappears for a few frames because earlier motion remains available [7, 8, 9]. That temporal information distinguishes it from estimating a removed ball in one edited image.

Maksai et al. show how player information can support ball localisation by combining ball–player interactions with physical constraints on the trajectory [7]. Calibrated cameras and changes between states such as possession and flight help track the ball when detections are weak or missing. Without that temporal and three-dimensional information, the present system instead selects one player and learns an image-coordinate offset from that player.

A different approach is to learn a ball heatmap directly from image pixels, as Huang et al. do in TrackNet [8]. They examine single-frame and consecutive-frame inputs, with consecutive frames adding motion information when the ball

is small, blurred or difficult to see. This offers an alternative to extracting player measurements first, although its published tennis-video scores are not a baseline for edited football frames. Training an image-to-ball model on the same football data would give a closer comparison.

The difficulty of football tracking is reflected in SoccerNet-Tracking, which provides bounding boxes and identities for players, referees and the ball across video sequences [9]. Cioppa et al. identify fast motion and severe occlusion as difficult conditions. These motivate using information beyond a visible ball, whether to follow it through time or, as here, to estimate one missing location from a single image.

3.3 Gaze direction and image locations

A person's looking direction may point towards an object, making gaze a possible clue to a missing ball. A direction still leaves the target distance unkowns, so turning it into an image location requires assumptions about the camera, scene and distance.

L2CS-Net uses a face image to predict yaw and pitch separately, describing horizontal and vertical gaze direction [10]. Its training combines broad angle categories with a finer numerical estimate. A football predictor can use those angles, but must still choose a distance along the predicted direction.

Head orientation provides a related, rougher clue. Hempel et al.'s 6DRepNet estimates three-dimensional head rotation from appearance [11], which describes where the head faces. Because the eyes can move while the head stays still, its usefulness as a gaze estimate depends on how closely those directions agree.

Gaze360 provides labelled three-dimensional gaze data across varied indoor and outdoor settings, head poses and distances [12]. Its method uses several frames and estimates uncertainty. The change to single football frames and a different recording setup means accuracy here needs a separate check.

3.4 Gaze following from scene information

Scene-based gaze following combines a person's head position with their surroundings to predict a target location. In GazeFollow, introduced by Recasens et al., the head position identifies whose gaze is being considered and the image provides possible targets [13]. Together, they support a prediction of where that person appears to be looking within the image.

Gaze-LLE simplifies the learned components of this approach [2]. Ryan et al. keep a pretrained DINOv2 image encoder unchanged and train a smaller decoder

to turn its visual features into a gaze heatmap, using a head-position marker to identify the relevant person.

The unmarked setting is relevant here because Ryan et al. also observe plausible targets without supplying a head position, including effective predictions in single-person examples. A marker is still needed to specify whose gaze is being estimated when several people are present [2]. The official implementation supports unmarked input, allowing this study to take the strongest heatmap location as a ball-position proposal and score it against the recorded position without first choosing a player. Person-marked outputs are compared separately in Section 5.6.3 and Appendix A.4.

3.5 Joint attention and player pose

When two or more people look towards a shared target, their directions can be considered together. Harari et al. estimate this joint attention using face detection, three-dimensional gaze and scene-depth estimates [14]. Depth helps distinguish objects that lie along the same line in an image, and their use of separately trained components also relates to the earlier football heatmap tool.

Relationships between people are central to Nakatani et al.'s method, which combines locations, actions and gaze with an image-based attention estimate [15]. Their experiments include volleyball, using the ball centre as the joint-attention location. Choosing one player keeps the football tool's calculation straightforward, but leaves out this agreement between people's positions and actions, which may contain clues absent from a single player's measurements.

Ultralytics provides YOLOv8 pose models that detect people and estimate landmarks such as shoulders, hips, knees and ankles [16]. Their output also includes facial points, with coordinates and estimated visibility for each landmark. These features allow the football location model to use a player's visible body arrangement when predicting the ball. Although the points include parts of the face, they do not measure where the player's eyes are looking.

3.6 Player selection and location models

Recorded ball locations supply labels for both player selection and location prediction. The study implements these separate supervised models using scikit-learn [17].

Logistic regression scores the candidate players [18], after which gradient-boosted regression trees predict horizontal and vertical offsets from the chosen player's centre. Both stages can use position features or the fuller representation

containing pose and box shape, allowing those inputs to be compared through the same prediction path.

3.7 Football data and image editing

The ball annotations released with SoccerNet-v3D provide a target for missing-ball prediction [4]. Gutiérrez-Pérez and Agudo extend football broadcast data with camera calibration and corresponding replay views for three-dimensional scene understanding and ball localisation. This study uses the released image dimensions and ball-box centres to evaluate two-dimensional locations, leaving three-dimensional reconstruction outside its scope.

Telea's inpainting method fills a selected region with a plausible continuation of its surroundings [5]. The OpenCV implementation creates the missing-ball inputs used here, giving every method the same edited image and retaining the true ball position for scoring. Traces from that edit or changes to nearby content mean artificial removal may differ from natural occlusion.

An editing trace could become a shortcut: a convenient pattern that supports prediction but may fail under other conditions, as described by Geirhos et al. [19]. A strong score would then give an incomplete account of what the model had learned to use. Lapuschkin et al. investigated such clues by changing visible source marks in images and checking the predictions [20], illustrating how a controlled input change can help examine them.

For the football images, adding a distant patch offers a small check of whether prediction is drawn towards an edit. Its placement matters because it can also remove useful scene content, especially when it overlaps a player. This tests one response to editing, while the broader question of which scene features support localisation remains open.

3.8 Evaluation and the role of simple guesses

Evaluation measures can lead to different views of performance. Bylinskii et al. show this for visual saliency, where scores differ in their treatment of location errors and central preferences [21]. The same need to examine more than one property applies here: mean distance reflects large misses, while the median and close count help show whether a method also gives accurate estimates on some images.

Simple baselines help separate those scores from the information a model actually uses. Centre and the training-position mean do not interpret the current image, while comparing full and position-only features examines what the extra measurements add beyond box and detection information.

The data split affects the meaning of a score too. Kapoor and Narayanan discuss how information leakage can make machine learning results look better than they are [3]. Frames from the same match may share information that makes prediction easier, so match membership matters when separating training and evaluation. Chapter 2 describes that separation alongside the earlier development exposure.

Bootstrap intervals add the scores by showing how differences vary across resampled data [6]. Resampling whole matches keeps related frames together, and using the same draws for each method preserves the pairing needed to compare their errors.

Chapter 4

System Design

4.1 Purpose and processing path

The tool predicts where a removed football had been by selecting a detected player and estimating the horizontal and vertical distance from that player's centre to the ball.

Figure 4.1 shows the route through the tool. After player detection and feature extraction, the system selects one player and calculates a raw ball coordinate. It also produces a blended prediction by moving that point towards the average training-ball position. Both predictions are checked in the evaluation. Chapter 2 defines the comparison, while this chapter explains how the software produces them.

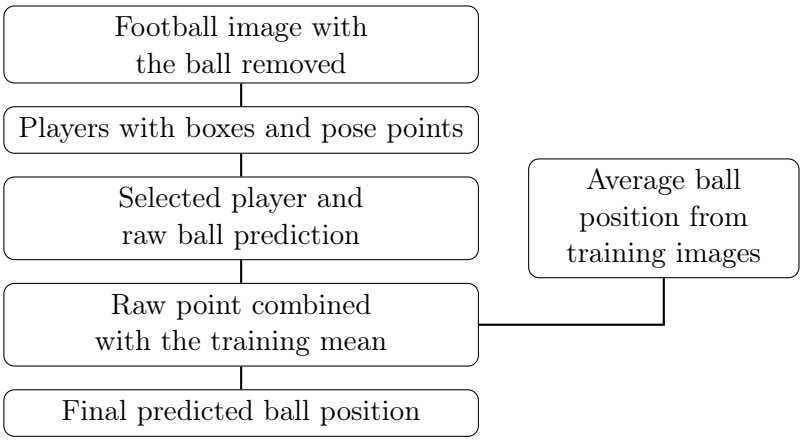


Figure 4.1: Main player-based prediction path, read from top to bottom. The system uses the training-position mean when no player remains after filtering.

4.2 The earlier head and eye approach

Development began with the directions in which players appeared to be looking. YOLO detected people, whose boxes were matched to pose detections so visible nose, eye and ear landmarks could locate their heads and provide face crops. Matching kept each estimate connected to the correct person, since overlapping player boxes could otherwise receive the same head.

L2CS-Net estimated horizontal and vertical gaze angles from the face crops using its existing Gaze360 weights [10, 12], while 6DRepNet estimated head orientation [11]. The two estimates describe related clues, but a player can move their eyes without turning their head, so the directions can differ.

To turn a selected direction into a heatmap, the tool drew a widening band from the player's head. Each band assigned support around its estimated direction, allowing some uncertainty in the angle. Visibility and image-quality measurements controlled each player's contribution. Overlapping bands gave higher scores where several players' estimated directions agreed.

The tool normally used L2CS. When its unit-length gaze direction points almost straight towards or away from the camera, only a short component runs across the image. If that on-image length was below 0.25, the tool switched to head orientation only when the head direction's on-image length reached the cut-off. Otherwise, it kept L2CS. If L2CS produced no estimate at all, it used the available head-orientation estimate.

Combining directions gives a heatmap, but several locations can still lie along the same line because direction alone does not give target distance. Small or partly hidden faces add difficulty to the angle estimate. Later checks covered the direction conventions and tested one combination with Gaze-LLE's heatmap. Chapter 5 reports this earlier work. The final player-based system takes a different approach: it learns the ball's position relative to a player directly from that player's measurements.

4.3 Image preparation

Testing the missing-ball task first requires hiding a ball whose position is known. Preparation reads the original image and its labelled ball box, rounds the box outwards and extends it by two pixels on each side. Any part outside the image is clipped at the boundary. Telea inpainting then fills this region using surrounding image information [5], with a radius of three pixels.

The labelled coordinate marks where to remove the ball. Prediction then uses the edited image without that coordinate, and scoring compares its output with the true ball position. This separates creating a missing-ball input from testing

what the remaining image reveals.

Checks compare the saved file with the prepared pixels and confirm that the edit has not changed anything outside its removal mask. Training and evaluation images follow the same editing rule, and the original evaluation files match the scene-model inputs exactly.

All image-based methods therefore start from the same edited image, although each can use it differently. Centre and the training mean simply return fixed points.

4.4 Player detection

YOLOv8m-pose detects the players using its pretrained weights [16]. For each detection it returns a bounding box, detection score and 17 estimated body and facial points, including shoulders, knees, eyes and ears. These describe the player's location, size and visible body arrangement for the later models.

The detector runs on the whole image and on four overlapping sections. It uses input size 1,280 for the whole image and 640 for each section, with 15% overlap. The sections give additional local views of players, including those near their boundaries. Running several passes can detect the same player more than once, so a merging step combines duplicate boxes. These sizes are detector settings, since the source images themselves have varying dimensions.

A field filter checks the proportion of pixels meeting a green-colour rule near the bottom of each candidate box, accepting those with enough green support. This helps reduce detections outside the playing area, although pitch appearance, clothing and shadows can affect a basic colour check.

Detection and field filtering both use the edited image, with controlled tests checking that each operation receives the expected colour-channel. A red/blue reversal can still be accepted by the software even though the colours are wrong. Chapter 5 describes the corrections, and Appendix A gives the detailed settings.

4.5 Player features

Feature extraction turns each accepted player's detections into numerical measurements: where the player stands, how large the box is and where the body landmarks lie. The position-only version records the box centre, width, height and area relative to the image, together with the detection and field-support scores. These seven features describe location and detection quality. Their comparison with the full feature set tests whether extra body information helps.

The full version adds box shape, the fraction of accepted pose points and the

locations of all 17 landmarks. Each landmark is recorded relative to the player's own box: a shoulder's measurement describes where it lies within that box. A similar body arrangement can then be represented on either side of the image. The box measurements still record the player's overall position.

Some landmarks are too uncertain to use. A point counts as visible when its confidence reaches 0.25. Below that threshold, the system sets its coordinate features to zero and records the missing measurement in a visibility flag. Without the flag, a model could confuse a missing point with an accepted point whose relative coordinate happens to be zero. This handling gives the models the same set of inputs even when only part of a player's pose is available.

There are 60 features in the full representation: the seven position values, two additional summary values and three values for each landmark. Those three are horizontal position, vertical position and visibility. Both versions use the same detected players, so their comparison concerns what the models can learn from the different feature sets, with the detector held unchanged.

4.6 Player selection

The nearest-box assignment in Chapter 2 supplies a training label for each qualifying image: the associated player is positive and the other accepted players are negative. Their features train the selector to score new candidates, using the known ball coordinates to make the training association consistent.

The logistic-regression selector uses standardised features [17], with training means and spreads also applied to evaluation inputs. This puts box area, coordinates and detector confidence on comparable scales. Balanced class weighting gives the less frequent positive examples more influence because negative player examples are much more prevelant.

The position-only and full versions have separate trained selectors. At prediction time, each scores the accepted players and selects the highest-scoring one, using the earliest player index to settle equal scores. The selector receives features from the edited image, without the true ball position. Its selected player becomes the reference for estimating the location.

Using one player gives the next stage a single starting point, instead of requiring it to combine a varying number of players. It receives that player's features and predicts a location relative to their centre. Bad selection can affect everything that follows. The highest selection score dont need to identify the most useful player in a crowded image. The answer-assisted check investigates this possibility with the same trained location models.

4.7 Predicting the ball position

Once a player has been selected, two regression models predict the horizontal and vertical offsets from the centre of that player's box. Their training targets are calculated by subtracting the associated player's centre from the labelled ball coordinate. A negative horizontal value means left and a positive value means right. Adding the predicted offsets back to the box centre produces the ball estimate.

Both location models use histogram-based gradient-boosted trees [18]. They learn the horizontal and vertical offsets from the player's features. The models can use position in both versions, with additional shape and landmark information available in the full version.

The offset models learn from one associated player in each qualifying frame, while the selector learns from all the player rows in those frames. Evaluation is broader: whenever accepted players are available, selection and location prediction run. Some evaluation images have no player close enough to the labelled ball to meet the training-distance rule. The eligibility check examines how the system behaves on those images as well as on images meeting the training rule.

The models and settings came from the earlier implementation. The models were trained once on the training set that excludes the evaluation matches, and the additional checks reused them. Appendix A records the settings needed to repeat the training procedure.

4.8 Raw and blended predictions

Adding the offsets to the selected player's centre gives the raw point. The system clips each coordinate between zero and one to keep it inside the image, then calculates a second prediction using the one-third blending rule. This combines the raw estimate with the average ball position from the training images. If $\mathbf{p}_{\text{raw}}$ is the raw point and $\mathbf{p}_{\text{mean}}$ is the training mean, the blended point is

$$\mathbf{p}_{\text{blend}} = \mathbf{p}_{\text{mean}} + \frac{\mathbf{p}_{\text{raw}} - \mathbf{p}_{\text{mean}}}{3}. \tag{4.1}$$

The blend lies one third of the way from the training mean towards the raw estimate. Because the raw coordinates are clipped to the image boundaries first, this also limits where the blended point can go. With the recorded training mean, its horizontal coordinate must lie between about 0.330 and 0.663, and its vertical coordinate between 0.327 and 0.660. The possible points occupy a rectangle one third of the image width and height, covering about 11.1% of its area. Even a correct raw estimate outside this rectangle is pulled back towards the middle.

The one-third weight was kept from the earlier implementation without validation tuning, so its value was a design choice rather than an established optimum.

The full blend remains the original main method, with the raw prediction reported alongside it. If filtering finds no usable player, both return the training-position mean.

4.9 Controls and saved outputs

The shuffled extra-description control exchanges box shape, landmark visibility and landmark coordinates between players. These features move together, while the seven position features stay attached to their original boxes. Both selection and offset prediction receive the changed features. Some combinations may describe no real player: a box's width and height can remain in place while its shape measurement comes from someone else. This check shows what happens when the extra features no longer belong to the player being described. The separately trained position-only model gives a more direct comparison of using the full feature set.

The exchange follows a repeatable ordering, leaving a single player's features unchanged because there is nobody to swap with. Empty detections use the usual fallback.

The program saves each image's player features, selection scores, selected player and predicted points. Together with the controls, these records allow the evaluation to follow an unexpected result back through detection, player selection and offset prediction.

Chapter 5

System Evaluation

5.1 The comparison

The evaluation compares the player-based system with simple position guesses and Gaze-LLE, then examines why its predictions range from close estimates to large misses.

Results begin with the original sample of 61 images from 61 matches. The full blend is the main method. Raw predictions are also reported to show the effect of pulling estimates towards the training-position mean.

A second comparison uses all available images from those matches, with no change to the trained models or original predictions. These extra views give a fuller account of the pattern in the main result.

5.2 Results on the two samples

5.2.1 The original 61 images

Table 5.1 compares the full and raw predictions with the position-only and shuffled versions, the two fixed-position guesses and Gaze-LLE. All methods use the same images.

The full blend has lower average error than centre and the training-position mean, but Gaze-LLE performs best on all three measures. The raw model gets close more often than the fixed guesses, yet its average error is near the centre result.

The training-position mean is approximately $(0.4947, 0.4907)$, so its closeness to image centre explains their similar scores. Both check whether player information adds more than the general tendency for play to appear near the middle of the image.

Method	Mean error	Median	Close
Full blend	0.2131	0.2012	6/61
Raw prediction	0.2285	0.1681	21/61
Position-only blend	0.2132	0.2076	7/61
Shuffled-description blend	0.2227	0.2186	6/61
Training-position mean	0.2291	0.2334	4/61
Image centre	0.2302	0.2307	5/61
Gaze-LLE, no supplied head marker	0.1625	0.1284	24/61

Table 5.1: Results on the same 61 images. Error is distance in separately normalised image coordinates. Close means error at most 0.10.

The paired intervals include zero for the small gains over centre and the training mean, leaving those improvements unclear on this sample. The comparison with Gaze-LLE favours the pretrained model. Appendix Table A.1 gives the five original comparisons.

5.2.2 Additional images from the same matches

The expanded comparison uses all 813 available images from the excluded matches, adding 752 to the original sample. The input audit found 810 images in the source release's test list and three in its training list. None of their matches were used to train this study's models.

A single image may be unusually easy or difficult. The extra frames help show whether the first result depends heavily on that view, using the same trained models.

Matches with more available images should not dominate the comparison. Average error and close rate are therefore calculated within each match first, then averaged across the 61 matches. The first two result columns in Table 5.2 use this equal match weighting. The last column gives the actual number of close predictions across all 813 images. Dividing that count by 813 gives the separate image-weighted rate.

With more views from each match, the comparison is less dependent on a single image, and the paired intervals support a modest gain for the full blend over centre and the training mean.

Gaze-LLE remains stronger overall, while the raw method again combines many close estimates with larger mistakes that keep its average error near the centre result. Appendix Table A.2 gives the paired comparisons.

Method	Match mean	Match close	Close count
Full blend	0.2125	15.6%	125/813
Raw prediction	0.2294	32.3%	271/813
Position-only blend	0.2238	12.9%	108/813
Shuffled-description blend	0.2360	12.0%	98/813
Training-position mean	0.2260	10.5%	87/813
Image centre	0.2267	10.3%	87/813
Gaze-LLE	0.1909	34.7%	288/813

Table 5.2: Seven methods on 813 images from 61 matches. Match mean error and match close rate give each match equal weight. Close count gives the total across all images. All 16 player-detection fallbacks remain included.

5.3 Contribution of the player features

The position-only blend tests whether body and facial landmarks and the extra box measurements add useful information beyond player location and basic detection data. Its mean error is almost the same as the full blend on the original sample, and both this comparison and the shuffled-feature comparison have intervals allowing either version to perform better. Neither gives clear evidence of a benefit from the extra measurements on that sample.

On the wider sample, the full blend has lower average error than both the position-only blend and the shuffled-description blend. The paired intervals support those differences, as shown in Appendix Table A.2.

Its extra measurements include box shape, landmark visibility and facial and body points, so the gain cannot be attributed to body pose alone. The position-only comparison carries more weight here because shuffling can create inconsistent combinations of player features.

5.4 Close estimates and larger errors

The raw model makes more close predictions than centre: 21 images compared with five on the original sample, and its lower median also shows that it can get much nearer to the ball in some images.

The problem is what happens on the other images. Large misses leave its average error near the centre result, with no clear difference between them. The close count records predictions within the stated distance threshold, while average distance also includes the cost of the much poorer ones. Appendix Table A.3 gives both comparisons.

Figure 5.1 shows the full spread of errors. At each distance, a higher curve means that more images have an error at most that distance. This makes both

the raw model's close estimates and its large misses visible in the same plot.

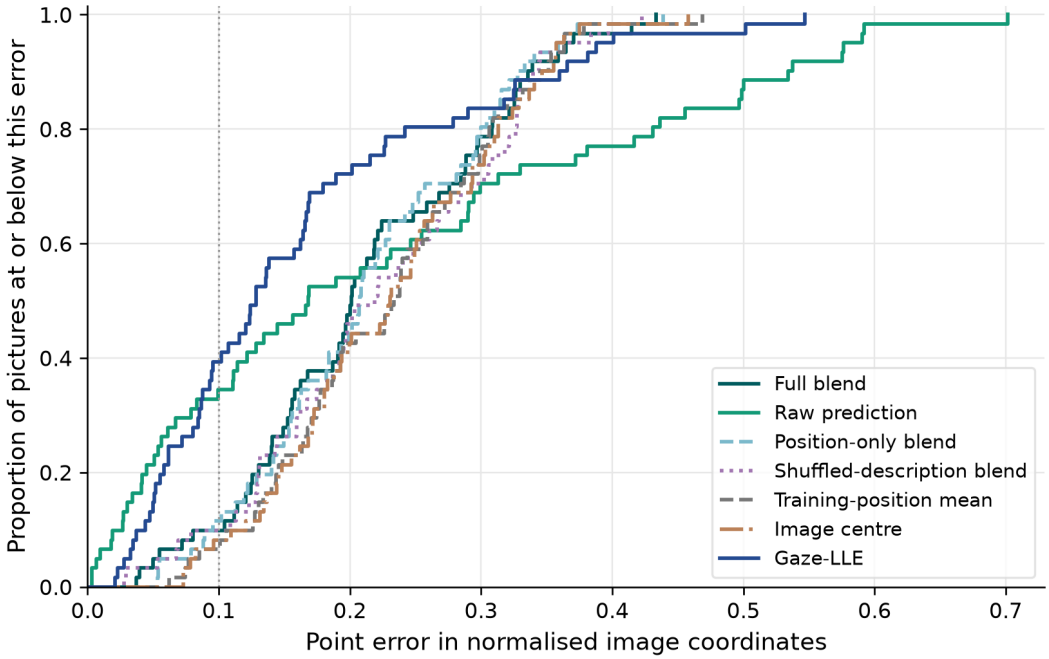


Figure 5.1: Fraction of the original 61 images with error at most each distance. The vertical marker shows the fixed 0.10 close threshold. Each curve uses the saved errors directly.

Blending pulls every raw point towards the same typical location, which can bring a very wrong estimate closer but also moves some good guesses away from the ball. In the original comparison, it lowers the close count from 21 to six. Average error falls, although the interval leaves that change uncertain. The blend has no way to tell which raw guesses are already good because it treats every point in the same way.

The recorded coordinates in Figure 5.2 make this effect easier to see. The first example is the earliest sample identifier where the raw point is close and the blend is not. Blending moves a raw estimate with error 0.0095 to one with error 0.2132. The second is the earliest identifier with raw error at least 0.40. Both diagrams show the selected player and estimates in relation to the ball. They use the recorded boxes and points on a blank background.

Gaze-LLE has a similar close count but substantially lower average error than the raw method. Its stronger overall result shows why close guesses alone are not

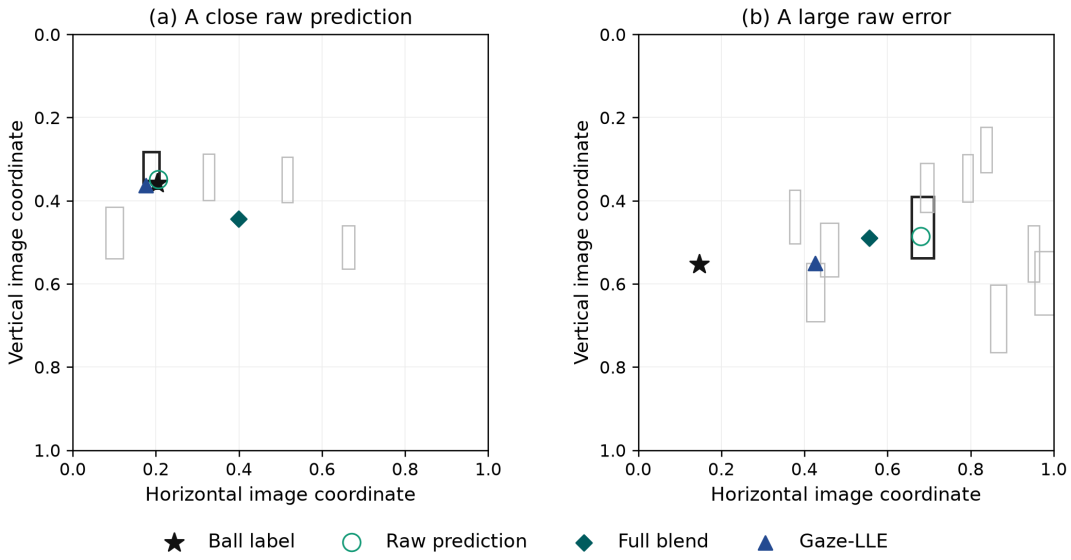


Figure 5.2: Recorded player boxes and ball-location estimates for a close raw prediction moved away by blending and a large raw error. The dark rectangle is the selected player and the pale rectangles are the other retained players. The examples follow fixed selection rules and illustrate those conditions.

enough. The expanded comparison shows the same broad pattern, making the larger errors the focus of the next checks.

5.5 Understanding the raw model's errors

5.5.1 Training-label eligibility

Since the offset models learned horizontal and vertical displacements from training images with a player box near the ball, this diagnostic checks what happens when that condition is absent. Using the known ball labels, an image meets the training rule when the ball is at most five per cent of the image diagonal from the nearest retained player box. This divides the sample into 37 eligible images and 24 ineligible images, including the no-player fallback.

	Eligible, 37		Ineligible, 24	
Method	Mean error	Close	Mean error	Close
Raw prediction	0.1465	20	0.3548	1
Full blend	0.1796	4	0.2648	2
Gaze-LLE	0.1428	17	0.1929	7

Table 5.3: The original images grouped by their labelled ball's distance from a retained player box. Each method uses the same images within each group.

The raw method gets close on 20 of the 37 eligible images and one of the 24 ineligible images, with much lower mean error in the eligible group. Gaze-LLE has similar observed mean error within that group and lower error in the ineligible group.

Blending raises average error in the eligible group and lowers it in the harder group, helping explain the trade-off. Moving towards a typical position reduces some large errors in images that do not meet the training rule, while losing close estimates in images that do.

This reveals a mismatch between the training set and the images on which a prediction is needed. Some evaluation balls are farther from every detected player box than the training rule permits. These cases remain in the evaluation and expose a weakness in the training coverage, although other differences between the groups may also affect difficulty.

5.5.2 Player selection

The next check uses answer-assisted selection to choose the nearest player box under the training-label proximity rule. The full selector agrees with it in 26 of 60 images with retained players and 22 of the 37 eligible images.

The chosen player's features enter the full offset regressors with clipping and the one-third blend unchanged, and all original images are scored. The choice follows box proximity rather than searching for the player whose predicted ball point is closest to the true ball position.

Selection and point rule	Mean error	Median	Close
Actual selector, raw	0.2285	0.1681	21/61
Answer-assisted selector, raw	0.1029	0.0796	34/61
Actual selector, blend	0.2131	0.2012	6/61
Answer-assisted selector, blend	0.1740	0.1634	8/61

Table 5.4: Offset models with actual or answer-assisted nearest-box selection on all 61 images. The assisted choices require the ground-truth ball positions.

Answer-assisted selection gives much lower raw error and more close predictions, but simply returning that player's centre gives mean error 0.1150 and the same close count of 34 as the offset model, whose mean error is 0.1029. Much of the assisted result therefore comes from the player choice, with a smaller additional improvement from the offsets. The assisted blend also lowers average error, although its small rise in close count remains uncertain. Appendix Table A.4 records the paired changes.

Player choice therefore helps explain the uneven raw results. Because a different player also supplies different features, the improvement reflects how selection and offsets work together.

5.5.3 What the offsets add

A supplementary comparison checks whether predicting an offset helps more than simply returning the selected player's centre. Table 5.5 uses the full selector's player for the centre guess and also shows raw position-only prediction.

	Original 61		All 813	
Point rule	Mean	Close	Match mean	Match close
Full raw prediction	0.2285	21/61	0.2294	32.3%
Position-only raw	0.2243	20/61	0.2693	24.2%
Chosen player's centre	0.2417	23/61	0.2441	27.3%

Table 5.5: Offset and player-centre comparisons. The 813-image columns give each match equal weight. The centre guess uses the full model's selected player. Position-only prediction has its own selector.

On the original 61 images, predicting an offset reduced average error compared with using the selected player's centre, but gave slightly fewer close predictions.

On the larger sample, it improved both measures. The difference between full raw prediction and position-only raw prediction was still unclear on the original sample. Appendix A.6 gives the paired comparisons.

The full and position-only models can choose different players, so their difference concerns the complete prediction path rather than the offset stage alone.

5.6 Supporting checks

5.6.1 Inputs and replay

Tests reproduced two input problems in an earlier version: the field filter reopened the image with the ball still visible, and the detector received colour channels in the wrong order. Both were corrected before producing the player-based comparison reported here, so detection and field filtering used the same ball-removed image as Gaze-LLE.

Replaying the predictions reproduced the points exactly in both samples. Appendix A records the preparation checks, duplicate-image findings and handling of images without detected players.

5.6.2 An additional distant edit

The editing check examines whether a trace left by removing the ball could attract Gaze-LLE's prediction. A second patch was edited into each original evaluation image using the same removal method and patch size. A recorded random rule placed it separately from the original mask and at least 0.20 from the ball in normalised image coordinates. The baseline rerun reproduced every saved heatmap before the added patches were tested.

All patches changed pixels, but sixty predicted points stayed unchanged, one moved away from the patch and none moved towards it. The close count stayed 24. Appendix A.3 gives the detailed results.

The added patch did not attract the strongest prediction in these images.

5.6.3 Older person-marked predictions

Older person-marked Gaze-LLE outputs are available for 49 of the same images. Each person's heatmap was normalised, then the maps were averaged before taking the strongest location. On these matched images, Gaze-LLE without a supplied head marker has slightly lower observed average error than this person-marked average. Both have lower average error than the full blend and raw prediction. Twelve person-marked outputs are unavailable, so this supporting comparison uses

only the common images. Appendix A.4 gives the scores separately from the complete sample.

5.6.4 Earlier head-and-eye heatmaps

An earlier check multiplied the head-and-eye heatmap with Gaze-LLE's scene heatmap using one fixed rule, after testing the direction convention. Multiplication gives high combined scores where both maps support the same location. The check then selected the highest-valued ten per cent of image area from each map and compared ball coverage in these equally sized regions. This measures region coverage rather than the single-point error used in the main evaluation, while giving neither map more area to cover the ball.

The combination included the ball in 39 of 61 images, against 46 for the scene map alone, so the tested combination gave no improvement.

5.7 Discussion

The comparisons show why the player-based system can give close estimates without maintaining a low average error. Holding the selected player unchanged isolates what the offsets add over that player's centre, while the selection and blending checks explain where larger changes in performance occur.

The centre baseline is important in this setting. The training-position mean is close to the centre, and both give similar average errors on the original sample. A prediction can therefore obtain a reasonable average distance without responding to the players. *Spot the Ball* identifies errors from centre guessing and choosing the wrong player as a reference [1]. These examples reinforce the value of the study's centre and player-centre guesses: a plausible-looking estimate may offer little beyond them.

The diagnostic checks point to player selection as a main source of large errors. Changing the player gives a much larger improvement than adding offsets to the player's centre under answer-assisted selection, making the reference player the main priority while showing that offsets contribute a smaller gain. Training coverage also leaves a gap because the models learn from nearby player–ball pairs but must predict when the ball is far from every player. The larger errors in that group show where this local training rule falls short. Maksai et al. use motion and physical constraints alongside player–ball interactions in multi-camera video [7], whereas a single-image system has no trajectories or calibrated views to help resolve the ambiguity.

The fixed blend is the main design weakness. Its one-third weight restricts predictions to approximately 11.1% of the image area, so even a good raw prediction

outside that region is pulled back towards the middle. Removing it preserves more close estimates but leaves the large misses exposed. A match-separated validation set would allow the weight to be chosen by testing this trade-off, instead of keeping the same correction for every prediction.

Gaze-LLE remains the stronger comparison method [2]. Its unmarked output gave better ball-location estimates than the player-based pipeline on these edited inputs, even without football fine-tuning. A model trained to predict the missing ball directly from pixels would provide a closer test of image-based learning against player features, since that comparison is absent here. The broader question of which representation works better therefore remains open.

The larger sample supports modest gains from offsets and fuller player features, using more views from the same 61 matches. The results point to three priorities: selecting a suitable player, covering harder ball locations during training and handling large misses.

5.8 Limitations

Artificial ball removal can leave traces or damage nearby features, so it differs from a ball naturally hidden behind a player or another object. The added-patch check tests one particular input change and does not establish whether the original removal patch influenced Gaze-LLE. Heatmaps may change without moving their strongest point, and the extra patch can remove useful scene information. Further evaluation should cover naturally hidden balls, other sports and camera settings. A separate user study would be needed to establish whether the predictions help in practice.

All 379 matches had been used during earlier development. Evaluation matches are excluded from fitting, but the model family, settings and blend come from that exploratory work. The intervals treat those choices as fixed [3] and apply to each comparison individually. Independent external data would be needed to confirm performance beyond these matches.

Preparation and training changed together between implementations, so their separate effects were not measured. Not every training image was manually inspected. Nearest-box labels describe proximity, not attention or player role.

Chapter 6

Conclusion

6.1 Answer to the research question

Player information can help locate a missing football, but the completed system is still uneven. On the original 61 images, the full blend did not show a clear mean-error improvement over the fixed-position baselines. However, the raw prediction produced more estimates within the close threshold than centre guessing. The expanded comparison on 813 images from the same 61 matches supports modest improvements over image centre and the position-only blend. Gaze-LLE remains stronger overall. These findings remain exploratory because the evaluation matches had already been used during earlier development.

The raw predictions show the main trade-off: some land close to the ball, but large misses leave average error near the centre baseline. Pulling them towards a typical position lowered average error while also moving some good guesses away from the ball, so that reduction came at the cost of fewer close estimates.

The diagnostic comparisons make the role of the learned offsets clearer. Predicting a location from the selected player improves average error compared with pointing at that player's centre. It does not improve the original close count, although both measures improve in the larger sample. Under answer-assisted selection, the centre and offset predictions give the same close count. Much of that diagnostic's accuracy comes from selecting a player near the known ball.

6.2 Contribution of the work

The project produced a ball-location system and used comparisons of its stages to explain what the player features contribute. Centre guessing and Gaze-LLE gave reference points for judging the complete method, while the player-centre control separated the value of the learned offsets from the value of the selected

player. This makes the contribution more specific than the final score alone: the evaluation identifies which decisions help and where the system loses accuracy.

The expanded comparison supports some value from the fuller player measurements, which include box shape and point visibility as well as body pose. It also shows the limits of the training rule: learning locations near a player does not cover every image on which a prediction is needed. Together with the answer-assisted check, this gives a reason to improve player selection and training coverage before adding more complexity to the prediction model.

The supporting checks also help interpret the alternatives. A distant inpainted patch did not attract Gaze-LLE's predicted point, although the original removal patch may still matter, and combining the head-and-eye estimate with the scene heatmap gave no improvement. These results keep the focus on the player-based system and the weaknesses identified by its evaluation.

6.3 Next steps

Player selection is the clearest next step because the answer-assisted comparison shows how much the reference player can change the result. The nearest box supplies a simple training label, but proximity may not identify the player whose information is most useful. Independent labels describing useful player roles would allow that assumption to be checked before building a new selector.

Changes to selection or blending would need separate development data and testing on new matches with checked ball locations. A validation set separated by match could be used to choose the blend weight by balancing average error against the loss of close estimates. This would address the observed weakness of keeping the same weight for every input.

A direct image-to-ball model trained on the same football data would give a closer baseline for the player-feature approach than the current scene comparison. Testing naturally occluded balls would then show whether the method extends beyond artificial removal, provided their hidden locations can be established for scoring.

These steps aim to reduce large misses without losing the close estimates the system already produces. Once that behaviour is more reliable, a user study could check whether suggesting a ball location actually helps someone annotate an image.

Appendix A

Implementation and supporting results

Project code: `https://github.com/Juice234/Quorum-Gaze`.

A.1 Settings needed to repeat the comparison

YOLOv8m-pose ran on the whole image at size 1,280 and four overlapping tiles at size 640. Tile overlap and detection confidence were both 0.15. Duplicate boxes were merged at intersection-over-union 0.45 or containment 0.75, with a score factor of 0.85 for tile-edge detections. The field-support cutoff was 0.15 and the keypoint-confidence cutoff was 0.25. Ball removal used outward-rounded boxes, two pixels of padding and Telea inpainting with radius three.

The selector used standardised features and logistic regression with balanced class weights, at most 2,000 iterations and seed 20260703. The offset regressors used histogram-based gradient boosting with learning rate 0.05, at most 250 iterations, 15 leaves per tree, at least 20 samples per leaf and regularisation 1.0. Horizontal seeds were 20260703 for position-only and 20260803 for full features; vertical seeds added 100. Other parameters used the library defaults.

The training mean was approximately $(0.4946991, 0.4906810)$. Raw coordinates were clipped to the image before the one-third blend. Images with no accepted player used the training mean and remained in the evaluation.

Gaze-LLE used the DINOv2 ViT-B/14 in/out checkpoint, trained on GazeFollow and fine-tuned on VideoAttentionTarget [2]. Inputs were resized to 448×448. The model returned a heatmap and in/out score; the native 64×64 map was resized to 128×128 for scoring. The predicted point was the strongest cell's centre, with ties settled by the first maximum in row order. The main comparison reused these saved outputs without a head marker.

The environment used Python 3.11.9, PyTorch 2.4.1, NumPy 2.4.6, scikit-learn 1.9.0 and OpenCV 4.10. Detection used CPU single precision, eight threads and seed 2026101502. Bootstrap intervals used 5,000 shared whole-match draws, NumPy RandomState seed 20260721 and linearly interpolated percentile limits.

Input checks confirmed that removal changed no pixels outside the mask. Two groups of repeated training images stayed in training, with none crossing into evaluation. Replayed predictions matched the saved coordinates.

A.2 Original paired comparisons

Table A.1 gives the five original comparisons from Chapter 5. Differences are full blend minus comparator, so negative values favour the blend. Intervals use the same sampled matches for both methods.

Comparator	Difference	Pointwise 95% interval
Training-position mean	−0.01601	[−0.03461, +0.00269]
Image centre	−0.01710	[−0.03539, +0.00140]
Position-only blend	−0.00011	[−0.01182, +0.01148]
Shuffled-description blend	−0.00960	[−0.02858, +0.01039]
Gaze-LLE	+0.05058	[+0.02122, +0.07872]

Table A.1: Full blend minus each comparator on the original 61 images. These exploratory intervals describe each comparison separately.

Expanded and supporting paired results

The following tables collect the detailed comparisons discussed in Chapter 5. They use the same saved results and resampling procedures as before. Negative error differences favour the first method, while positive close-rate differences favour it. Each interval is a pointwise 95% interval.

Comparator	Difference	95% interval
Image centre	−0.01415	[−0.01964, −0.00869]
Training-position mean	−0.01354	[−0.01922, −0.00795]
Position-only blend	−0.01130	[−0.01562, −0.00679]
Shuffled-description blend	−0.02352	[−0.02868, −0.01828]
Gaze-LLE	+0.02164	[+0.00907, +0.03411]

Table A.2: Full blend minus each comparator on all 813 images, giving each of the 61 matches equal weight. These are the expanded-sample results.

Comparator	Difference	95% interval
Image centre	−0.00176	[−0.05316, +0.04962]
Gaze-LLE	+0.06593	[+0.01511, +0.11737]

Comparator	Change, pp	95% interval, pp
Image centre	+26.23	[+13.11, +39.34]
Gaze-LLE	−4.92	[−19.67, +9.84]

Table A.3: Full raw prediction minus centre or Gaze-LLE on the original 61 images. Close-rate changes are percentage points (pp). The error and close-rate measures describe different parts of performance.

Comparison	Difference	95% interval
Assisted raw minus actual raw	−0.12560	[−0.17006, −0.08295]
Assisted blend minus actual blend	−0.03912	[−0.05536, −0.02311]

Comparison	Change, pp	95% interval, pp
Assisted raw minus actual raw	+21.31	[+11.48, +31.15]
Assisted blend minus actual blend	+3.28	[−3.28, +9.84]

Table A.4: Answer-assisted predictions minus their ordinary-selector counterparts on the original 61 images. The assisted choice requires the known ball location. Close changes are percentage points.

A.3 Details of the follow-up checks

Eligibility used the shortest ball-to-box distance divided by the image diagonal. Ties used box-centre distance, then player index. Answer-assisted selection used the same rule and full offset models. Images without players were ineligible and kept the training-mean fallback.

All 61 extra patches changed pixels only inside their masks. Eight overlapped a player box. This records box overlap, without a manual check of the content removed. The baseline rerun matched all 61 saved heatmaps. Mean ball error was 0.16254 before the extra patch and 0.16217 after it, with 24 close predictions in both cases.

A.4 The common 49-image comparison

Table A.5 compares every method on the same 49 images with person-marked outputs. The other twelve had no usable head output and remain in the full 61-image comparison without a head marker.

Method	Mean error	Median	Close
Full blend	0.2194	0.2002	3/49
Raw prediction	0.2262	0.1664	17/49
Position-only blend	0.2166	0.2076	5/49
Shuffled-description blend	0.2316	0.2216	4/49
Training-position mean	0.2345	0.2380	2/49
Image centre	0.2357	0.2316	2/49
Gaze-LLE, no supplied head marker	0.1700	0.1358	17/49
Gaze-LLE, person-marked average	0.1811	0.1230	17/49

Table A.5: Point-localisation results on the identical 49 images with earlier person-marked outputs. Close means error at most 0.10.

A.5 Data use

The images and ball annotations come from SoccerNet-v3D, and the physical ball-box centres provide the evaluation targets.

A.6 Supplementary player-selection comparisons

The position-only raw points were recovered by reversing the blend: training mean plus three times the difference between blend and mean. Reblending matched within 10^{-12}. No clipping was needed, and rounding did not affect the 0.10 close classification.

The centre controls used the full selector's player or the answer-assisted player. They kept the training-mean fallback for one original case and sixteen expanded cases. Full and position-only selectors chose different players in 341 images, so their comparison includes both selection and offset prediction.

Tables A.6 and A.7 give the five paired comparisons using the resampling procedure above.

First method minus comparator	Difference	95% interval
61: full raw minus position-only raw	+0.00414	[−0.02548, +0.03210]
61: full raw minus chosen centre	−0.01324	[−0.02184, −0.00469]
813: full raw minus position-only raw	−0.03986	[−0.05331, −0.02641]
813: full raw minus chosen centre	−0.01465	[−0.01713, −0.01219]
61: assisted raw minus assisted centre	−0.01213	[−0.01857, −0.00576]

Table A.6: All five supplementary mean-error comparisons. Negative differences favour the first method. The assisted row uses the ground-truth ball position to choose the same player for both methods.

First method minus comparator	Change, pp	95% interval, pp
61: full raw minus position-only raw	+1.64	[−8.20, +11.48]
61: full raw minus chosen centre	−3.28	[−8.20, 0.00]
813: full raw minus position-only raw	+8.16	[+5.12, +11.26]
813: full raw minus chosen centre	+5.04	[+3.21, +6.93]
61: assisted raw minus assisted centre	0.00	[−6.56, +6.56]

Table A.7: The same five comparisons for close rate, expressed in percentage points (pp). Positive changes favour the first method. Both answer-assisted methods have 34 close predictions, although they need not be close on the same images.

Bibliography

[1] Neha Balamurugan, Sarah Wu, Adam Chun, Gabe Gaw, Cristobal Eyzaguirre, and Tobias Gerstenberg. Spot the ball: A benchmark for visual social inference. arXiv:2511.00261, version 2, 2025.

[2] Fiona Ryan, Ajay Bati, Sangmin Lee, Daniel Bolya, Judy Hoffman, and James M. Rehg. Gaze-LLE: Gaze target estimation via large-scale learned encoders. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025.

[3] Sayash Kapoor and Arvind Narayanan. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 4(9):100804, 2023.

[4] Marc Gutiérrez-Pérez and Antonio Agudo. SoccerNet-v3D: Leveraging sports broadcast replays for 3D scene understanding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, 2025.

[5] Alexandru Telea. An image inpainting technique based on the fast marching method. *Journal of Graphics Tools*, 9(1):25–36, 2004.

[6] Bradley Efron. Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, 7(1):1–26, 1979.

[7] Andrii Maksai, Xinchao Wang, and Pascal Fua. What players do with the ball: A physically constrained interaction modeling. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 972–981, 2016.

[8] Yu-Chuan Huang, I-No Liao, Ching-Hsuan Chen, Tsi-Ui Ik, and Wen-Chih Peng. TrackNet: A deep learning network for tracking high-speed and tiny objects in sports applications. arXiv:1907.03698, 2019.

[9] Anthony Cioppa, Silvio Giancola, Adrien Deliège, Le Kang, Xin Zhou, Zhiyu Cheng, Bernard Ghanem, and Marc Van Droogenbroeck. SoccerNet-Tracking:

Multiple object tracking dataset and benchmark in soccer videos. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, pages 3491–3502, 2022.

[10] Ahmed A. Abdelrahman, Thorsten Hempel, Aly Khalifa, and Ayoub Al-Hamadi. L2CS-Net: Fine-grained gaze estimation in unconstrained environments. arXiv:2203.03339, 2022.

[11] Thorsten Hempel, Ahmed A. Abdelrahman, and Ayoub Al-Hamadi. 6D rotation representation for unconstrained head pose estimation. In *2022 IEEE International Conference on Image Processing (ICIP)*, pages 2496–2500, 2022.

[12] Petr Kellnhofer, Adria Recasens, Simon Stent, Wojciech Matusik, and Antonio Torralba. Gaze360: Physically unconstrained gaze estimation in the wild. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2019.

[13] Adria Recasens, Aditya Khosla, Carl Vondrick, and Antonio Torralba. Where are they looking? In *Advances in Neural Information Processing Systems*, volume 28, 2015.

[14] Daniel Harari, Joshua B. Tenenbaum, and Shimon Ullman. Discovery and usage of joint attention in images. arXiv:1804.04604, 2018.

[15] Chihiro Nakatani, Hiroaki Kawashima, and Norimichi Ukita. Interaction-aware joint attention estimation using people attributes. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 10224–10233, 2023.

[16] Ultralytics. YOLOv8 and pose estimation documentation. `https://docs.ultralytics.com/models/yolov8/`. Official software documentation. Pose outputs described at `https://docs.ultralytics.com/tasks/pose/`. Accessed 15 October 2026.

[17] Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, Jake Vanderplas, Alexandre Passos, David Cournapeau, Matthieu Brucher, Matthieu Perrot, and Édouard Duchesnay. Scikit-learn: Machine learning in python. *Journal of Machine Learning Research*, 12(85):2825–2830, 2011.

[18] Scikit-learn developers. User guide: Logistic regression and histogram-based gradient boosting. `https://scikit-learn.org/stable/user_guide.html`.

Official software documentation, sections 1.1.11 and 1.11.1.1. Accessed 15 October 2026.

[19] Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A. Wichmann. Shortcut learning in deep neural networks. arXiv:2004.07780, 2020.

[20] Sebastian Lapuschkin, Stephan Wäldchen, Alexander Binder, Grégoire Montavon, Wojciech Samek, and Klaus-Robert Müller. Unmasking Clever Hans predictors and assessing what machines really learn. *Nature Communications*, 10:1096, 2019.

[21] Zoya Bylinskii, Tilke Judd, Aude Oliva, Antonio Torralba, and Frédo Durand. What do different evaluation metrics tell us about saliency models? arXiv:1604.03605, 2016.